



A Matrix-Free GPU-Accelerated Topology Optimization Framework for Biodegradable Implants Using Coupled Reaction-Diffusion Corrosion and Bone Remodeling

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MOTIVATION & RESEARCH NOVELTY

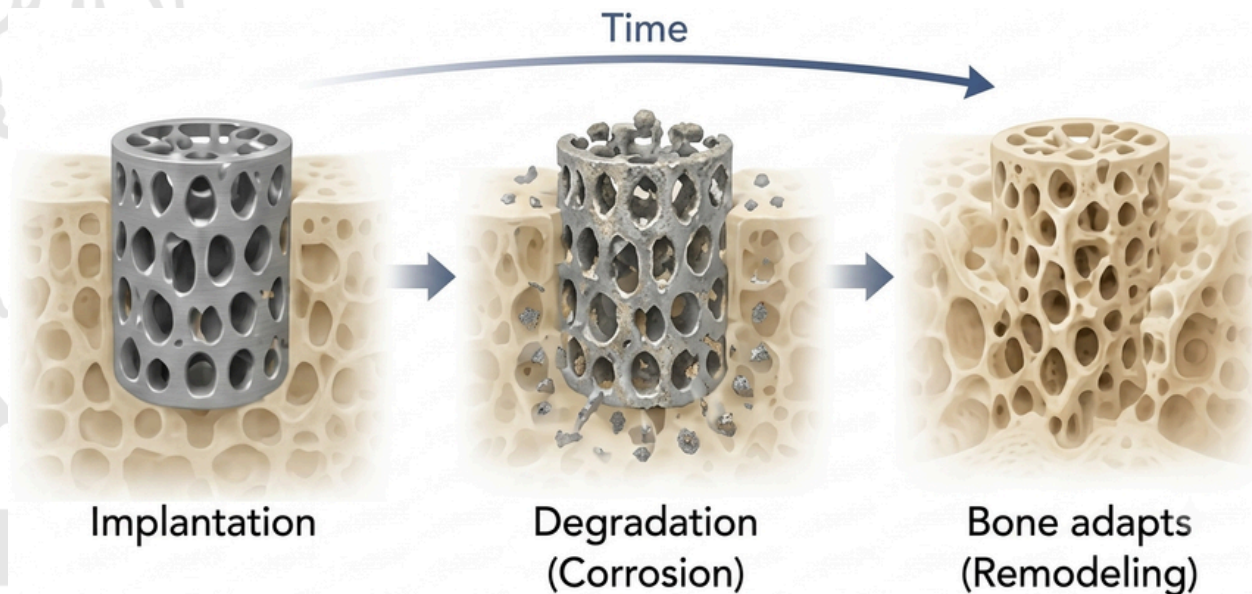
Biodegradable implants must perform over time and not just at implantation.

- Too-fast degradation → premature loss of mechanical integrity
- Too-slow degradation → prolonged stiffness & potential stress shielding
- Implant geometry evolves continuously due to corrosion
- Changing implant stiffness alters load transfer to bone
- Bone adapts/remodels to changing mechanical stimulus

Key Challenge

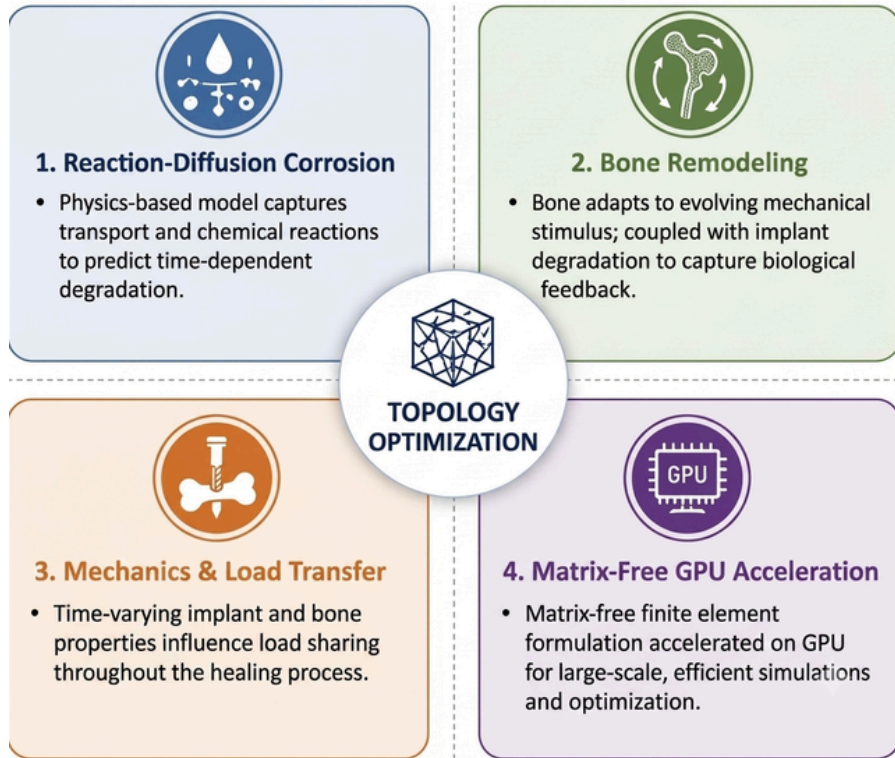
Design a biodegradable implant that supports the required mechanical load during healing and degrades at a controlled rate.

- The implant must retain sufficient strength and stiffness until the bone can support the load.
- Implant stiffness should decrease gradually so that load transfers to the healing bone without premature failure or stress shielding.



OBJECTIVES & MEMBER ROLES

HOW OUR WORK IS NOVEL?



OBJECTIVES

- 01** Optimize biodegradable implant topology for improved structural performance.
- 02** Predict and control time-dependent degradation while maintaining mechanical integrity throughout healing.
- 03** Develop an integrated and computationally scalable framework for degradation-aware implant optimization.

MEMBER ROLES

Aditi Jain

Corrosion & Degradation Modelling

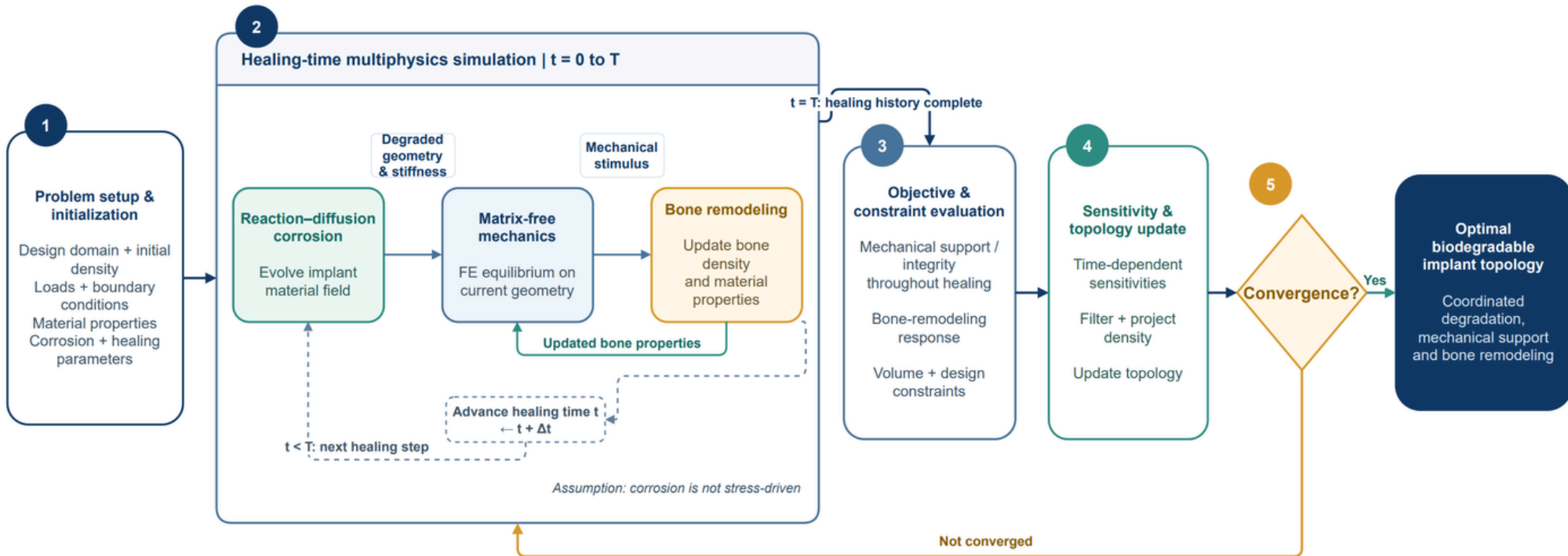
Garv Agarwal

Mechanics & GPU Implementation

Pratham Singla

Topology Optimization & Bone Remodeling

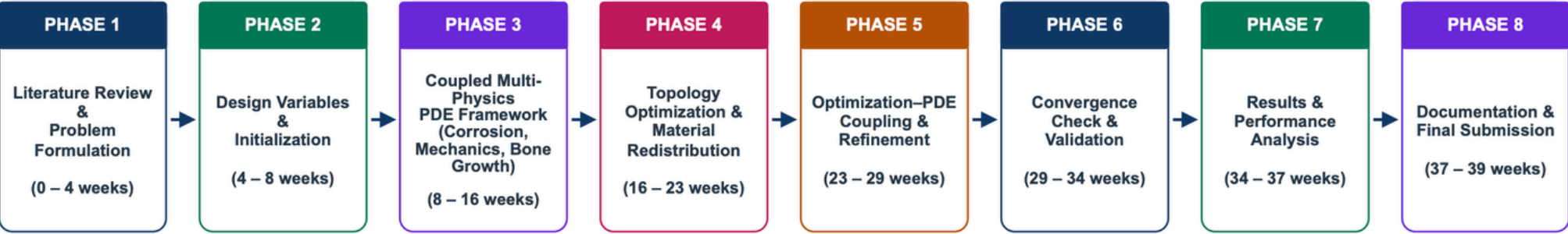
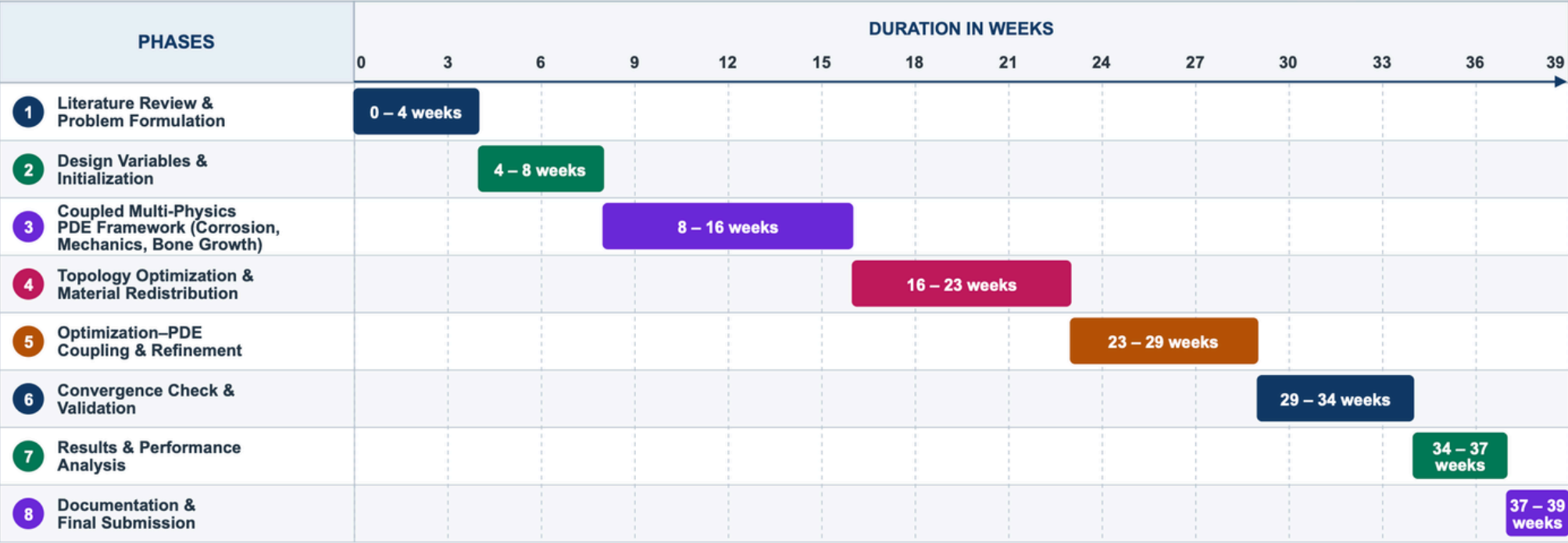
METHODOLOGY





ACTIVITY-TIME CHART

PROJECT TIMELINE & PHASES



Related Work & Research Gap

- Li et al. (2026), Computer Methods in Applied Mechanics and Engineering. This paper integrates phase-field corrosion directly into topology optimization. Their method studies how the structure loses material and stiffness over time, then improves the design so that it can continue providing mechanical support after degradation. This is the closest existing study to our main optimization idea.
- Sun et al. (2024), Journal of Biomedical Engineering. This study designed a biodegradable fixation plate whose stiffness decreases as the material degrades. They then studied how this changing stiffness affects load transfer and bone remodeling. The paper supports our goal of matching implant degradation with bone healing, although remodeling was evaluated only after optimization.
- Kovačević et al. (2023), Acta Biomaterialia. This paper developed a model for magnesium corrosion that includes uniform corrosion, local pits, and the effect of mechanical loading. The model was compared with experimental data and helps predict how an implant's shape and mechanical strength may change during degradation.
- Zhao et al. (2024), Computer Methods in Applied Mechanics and Engineering. This work used a matrix-free GPU method to speed up large topology optimization problems. Their approach reduces the storage and computational cost of repeated mechanical simulations. This supports our plan for GPU acceleration, although their method was tested on composite structures rather than biodegradable implants.

Existing studies cover corrosion, mechanical performance, bone remodeling, topology optimization, and GPU computing separately. Our project aims to connect these parts in one framework for designing biodegradable implants that remain mechanically safe throughout healing.

Thank You!

ACTIVITY-TIME CHART

